# ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 25-Sep-2026

Site: JNNCE | Location: Karnataka, Shimoga,sagara | Grid-Tied Solar Engineering Dossier

| PLANT CAPACITY | ANNUAL YIELD | CAPITAL COST | PAYBACK PERIOD |
|---|---|---|---|
| 550W Mono | Specific Yield | 12-Item BOM | 25Y Lifecycle |
| **3.00 kW** | **4,757 kWh** | **Rs. 165,850** | **5Y 4M** |
| 3.30 kWp Installed | CUF 18.1% \| PR 75% | Rs. 1.66 Lakhs | Profit: Rs. 13.68L |

## [SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

| | |
|---|---|
| **Company / Customer Name** | JNNCE |
| **Contact Mobile / Phone** | +918073398259 |
| **Location / Address** | Karnataka, Shimoga,sagara |
| **Contact Email Address** | adarshakmsgr50@gmail.com |
| **Design Path Selected** | Monthly Electricity Bill Units |
| **System Target Capacity** | 3.00 kW (Single LT Installation) |
| **Total Installed Array Rating** | 3.30 kWp |
| **Solar PV Module Rating** | 550W Mono PERC / TOPCon Half-Cut |
| **Total Module Count** | 6 Nos. |

## [METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

| | |
|---|---|
| **Total Rooftop Area** | 300.00 Sq.Feet (27.87 sq.m) |
| **Utilized Usable Rooftop Area** | 300 Sq.Feet (27.87 sq.m) |
| **First Year Annual Generation** | 4,756.62 kWh |
| **25-Year Cumulative Generation** | 105,681.90 kWh |
| **Estimated Payback Period** | 5 Years & 4 Months |
| **Post-Payback Profitability (25 Yrs)** | Rs. 1,368,329.33 (INR 13.68 Lakhs) |
| **Alternative FD 6.5% Benchmark (25 Yrs)** | Rs. 800,673.90 (INR 8.01 Lakhs) |

### ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 86 Tonnes CO2 | Tree Equivalence: 137 Mature Trees | Green Energy: 100% Clean LT Power

*Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.*

## [ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

| | |
|---|---|
| **Building / Array Identification** | Building 1 (Rooftop Array) |
| **Gross Rooftop Area** | 300.00 Sq.Feet (27.87 sq.m) |
| **Utilized Effective Rooftop Area** | 300 Sq.Feet (27.87 sq.m) |
| **Building Plant Target Capacity** | 3.00 kW |
| **Solar PV Module Rating** | 550 W Mono-PERC (Mono-facial) |
| **Number of Solar Modules** | 6 Nos. |
| **String Sizing Architecture** | 18 - 20 Panels per MPPT String (Optimal Voltage Band) |
| **Selected Site Inverter Rating** | 3 kW, 230V 1-Ph MPPT |

## [INVERTER] Selected Grid-Tied Inverter Specifications (3 kW)

Core inverter rating (70% derated selection), MPPT topology & models

| | |
|---|---|
| **Selected Inverter Rating** | 3 kW Grid-Tied Inverter (Single-Phase) |
| **Certified Top-Tier OEM Models** | Growatt MIC, Luminous NXi 130 |
| **Derated AC Design Power (70% Factor)** | 2.10 kW AC (from 3.00 kWp DC) |
| **AC Output Voltage & Grid Phase** | 230V, 1-Phase, 50 Hz AC |
| **MPPT Operating Topology** | Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency) |
| **Enclosure & Grid Protection** | IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116) |
| **Inverter to ACDB Cable** | 3.5Cx4 Sq.mm |
| **DC Array Isolator Switch** | 25A, MCB Type, 2P 230V |

## [MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

| | |
|---|---|
| **Incomer Overcurrent Protection** | Thermal-Magnetic MCB / MCCB with Manual Isolation |
| **Breaker Current Rating** | 16 Amps |
| **Rated Inverter Operating Current** | 9.13 Amps |
| **Trip Curve & Breaking Capacity** | Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1) |
| **Surge Protection Device (SPD)** | Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling |

## [CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

| | |
|---|---|
| **Solar PV DC Array Cable** | 4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618) |
| **Inverter AC Output Cable** | 3.5Cx4 Sq.mm (Cu / XLPE / Armoured) |
| **ACDB to Utility Grid Cable** | 3.5Cx6 Sq.mm (Al / XLPE / Armoured) |
| **Earth Grounding Conductor** | 16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits |
| **Design Standard & Voltage Drop** | Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52 |

## [ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

| | |
|---|---|
| **ACDB Enclosure Specification** | IP65 Weatherproof Polycarbonate / Sheet Steel |
| **Internal Busbar Rating** | 25A, 230V, 2PN Aluminium |
| **Incomer & Outgoing Terminations** | Cable Glands with Compression Lugs & Dual Earthing |
| **Interconnection Distance** | 100.0 Meters (ACDB to Utility Net-Meter) |

## [BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

| Sl. | Component / Material Description | Engineering Specification | Qty | Amount (INR) |
|---|---|---|---|---|
| 01 | Solar PV Modules (Mono PERC / TC | 550W Mono PERC Half-Cut Cells (>21.3% Efficiency) | 6 Nos. | Rs. 60,000 |
| 02 | On-Grid Solar String Inverter(s) | Dual MPPT, IP65 (JNNCE: 3 kW) | 1 Set(s) | Rs. 42,000 |
| 03 | Rooftop mounting structure | HDGI, module-compatible | 1 Site(s) | Rs. 15,000 |
| 04 | DC solar cable | 4-6 sq.mm copper, PV-rated | 1 Site(s) | Rs. 2,800 |
| 05 | AC cable | 4-6 sq.mm copper | 1 Site(s) | Rs. 2,250 |
| 06 | MC4 connectors | PV-rated connector pairs | 1 Site(s) | Rs. 1,800 |
| 07 | DCDB | Isolator + fuse + Type-II DC SPD | 1 Site(s) | Rs. 5,500 |
| 08 | ACDB | 16A MCB + 25A RCCB + Type-II AC SPD | 1 Site(s) | Rs. 4,500 |
| 09 | Earthing kit | Earth electrodes & accessories (IS 3043 - 3 Pits) | 1 Site(s) | Rs. 6,000 |
| 10 | Lightning arrester | Standard rooftop PV type | 1 Site(s) | Rs. 2,500 |
| 11 | Installation accessories | Lugs, ferrules, clamps, conduit, glands, etc. | 1 Site(s) | Rs. 3,500 |
| 12 | Installation & commissioning | Labour and commissioning | 1 Site(s) | Rs. 20,000 |
| | | **TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)** | | **Rs. 165,850.00** |

## [UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

| | |
|---|---|
| **Application & Registration Fee** | Rs. 500 + GST (DISCOM portal filing) |
| **Facilitation & Technical Processing Fee** | Rs. 1,000 + GST (Feasibility clearance) |
| **PPA e-Sign Stamp Duty Charge** | Rs. 11.8 (Electronic stamping) |
| **Bidirectional Utility Net-Meter** | Rs. 4,500 (DISCOM meter testing/procurement) |
| **Refundable Security Deposit** | Rs. 3,000 (Refundable utility escrow) |

**TOTAL CAPITAL INVESTMENT REQUIRED**

Turnkey Plant BOM Cost: Rs. 165,850 | ESCOM Utility Budget: Rs. 9,012 | Grand Total Outlay: Rs. 174,862 (Rs. 1.75 Lakhs)

## [FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

| CAPITAL INVESTMENT (CAPEX) | PAYBACK PERIOD (RETURN) | 25Y NET SAVINGS (PROFIT) | FD 6.5% BENCHMARK (BENCHMARK) |
|---|---|---|---|
| **Rs. 165,850** | **5Y 4M** | **Rs. 13.68L** | **Rs. 8.01L** |
| 12-Item BOM Basis | Break-Even Year | Rs. 1,368,329 | Alternative Asset |

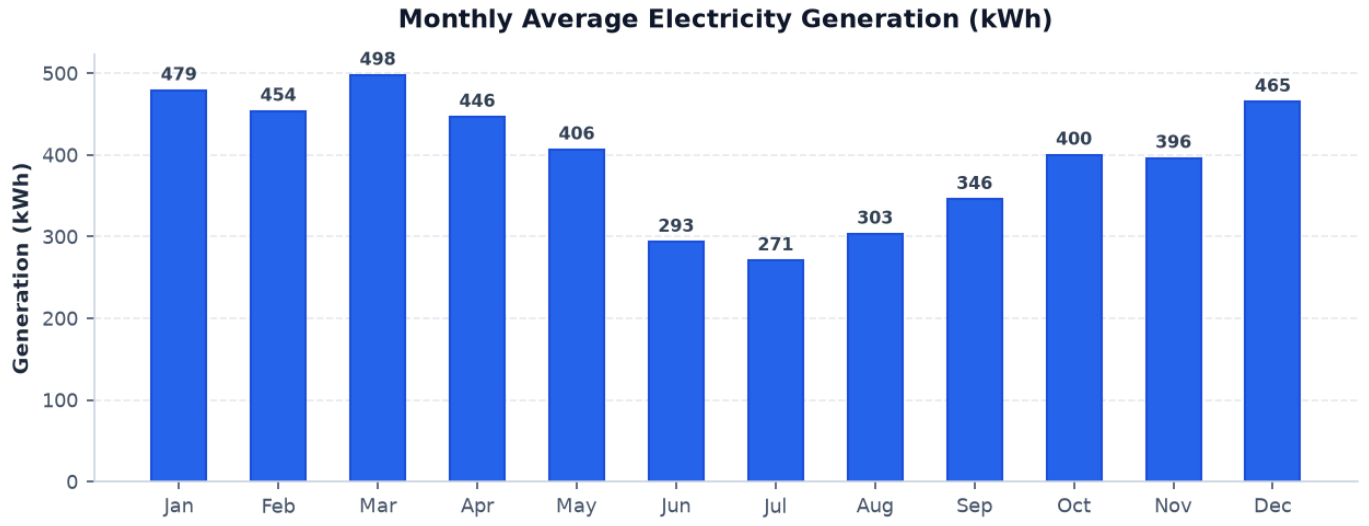
| Yr | Gen (kWh) | Tariff (Rs/u) | Gross Rev (Rs) | AMC (Rs) | Net Annual (Rs) | Cumulative Net (Rs) |
|---|---|---|---|---|---|---|
| 01 | 4,757 | Rs. 8.05 | 38,291 | 1,800 | 36,491 | 36,491 |
| 02 | 4,709 | Rs. 8.45 | 39,792 | 1,800 | 37,992 | 74,482 |
| 03 | 4,662 | Rs. 8.87 | 41,352 | 1,800 | 39,552 | 114,034 |
| 04 | 4,615 | Rs. 9.32 | 43,015 | 1,800 | 41,215 | 155,249 |
| **05 *** | **4,569** | **Rs. 9.78** | **44,687** | **1,800** | **42,887** | **198,136** |
| 06 | 4,523 | Rs. 10.27 | 46,456 | 1,800 | 44,656 | 242,792 |
| 07 | 4,478 | Rs. 10.78 | 48,276 | 1,800 | 46,476 | 289,268 |
| 08 | 4,433 | Rs. 11.32 | 50,187 | 1,800 | 48,387 | 337,655 |
| 09 | 4,389 | Rs. 11.89 | 52,187 | 1,800 | 50,387 | 388,042 |
| 10 | 4,345 | Rs. 12.48 | 54,229 | 1,800 | 52,429 | 440,470 |
| 11 | 4,302 | Rs. 13.11 | 56,397 | 1,800 | 54,597 | 495,067 |
| 12 | 4,259 | Rs. 13.76 | 58,601 | 1,800 | 56,801 | 551,868 |
| 13 | 4,216 | Rs. 14.45 | 60,924 | 1,800 | 59,124 | 610,992 |
| 14 | 4,174 | Rs. 15.18 | 63,362 | 1,800 | 61,562 | 672,554 |
| 15 | 4,132 | Rs. 15.93 | 65,827 | 1,800 | 64,027 | 736,581 |
| 16 | 4,091 | Rs. 16.74 | 68,483 | 1,800 | 66,683 | 803,264 |
| 17 | 4,050 | Rs. 17.57 | 71,160 | 1,800 | 69,360 | 872,624 |
| 18 | 4,010 | Rs. 18.45 | 73,976 | 1,800 | 72,176 | 944,800 |
| 19 | 3,969 | Rs. 19.37 | 76,889 | 1,800 | 75,089 | 1,019,888 |
| 20 | 3,930 | Rs. 20.34 | 79,932 | 1,800 | 78,132 | 1,098,020 |
| 21 | 3,890 | Rs. 21.36 | 83,100 | 1,800 | 81,300 | 1,179,320 |
| 22 | 3,852 | Rs. 22.43 | 86,391 | 1,800 | 84,591 | 1,263,911 |
| 23 | 3,813 | Rs. 23.55 | 89,797 | 1,800 | 87,997 | 1,351,909 |
| 24 | 3,775 | Rs. 24.73 | 93,354 | 1,800 | 91,554 | 1,443,462 |
| 25 | 3,737 | Rs. 25.96 | 97,017 | 1,800 | 95,217 | 1,538,679 |

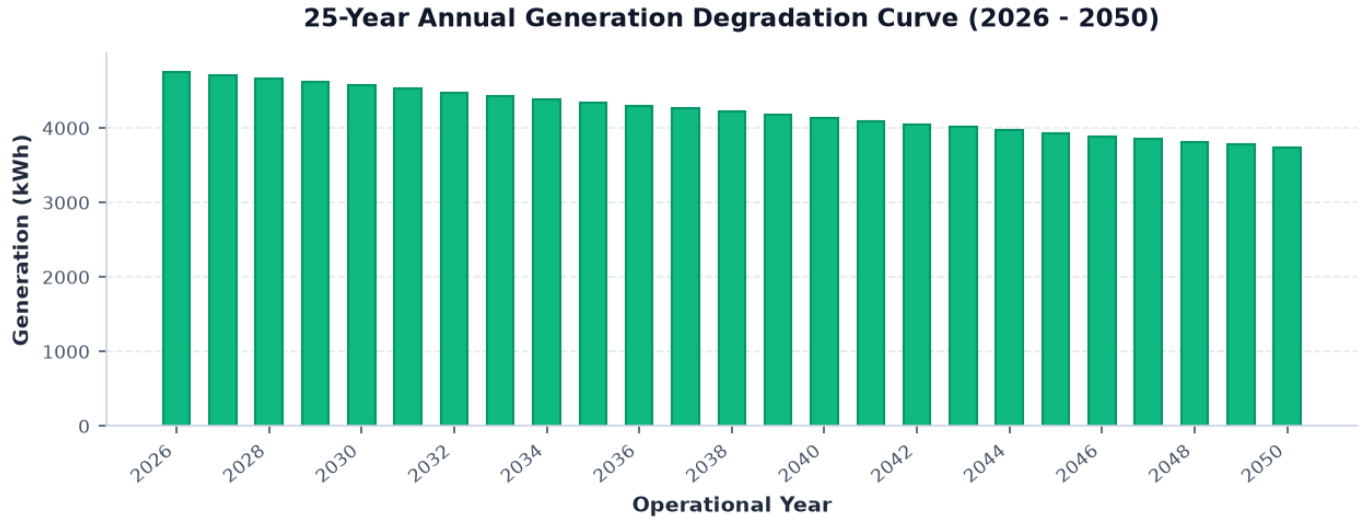
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

## [ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insulation yield and 25-year degradation profile

| SPECIFIC YIELD (YIELD) | PERFORMANCE RATIO (PR) | CAPACITY UTIL. (CUF) | DEGRADATION RATE (DEGRADATION) |
|---|---|---|---|
| **1586 kWh/kWp** | **75.0%** | **18.1%** | **1.0% / year** |
| kWh/kWp/Year | IEC 61724 Standard | Annual Plant Factor | Tier 1 Linear Warranty |




*Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.*

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# ON-GRID SOLAR PV SYSTEM

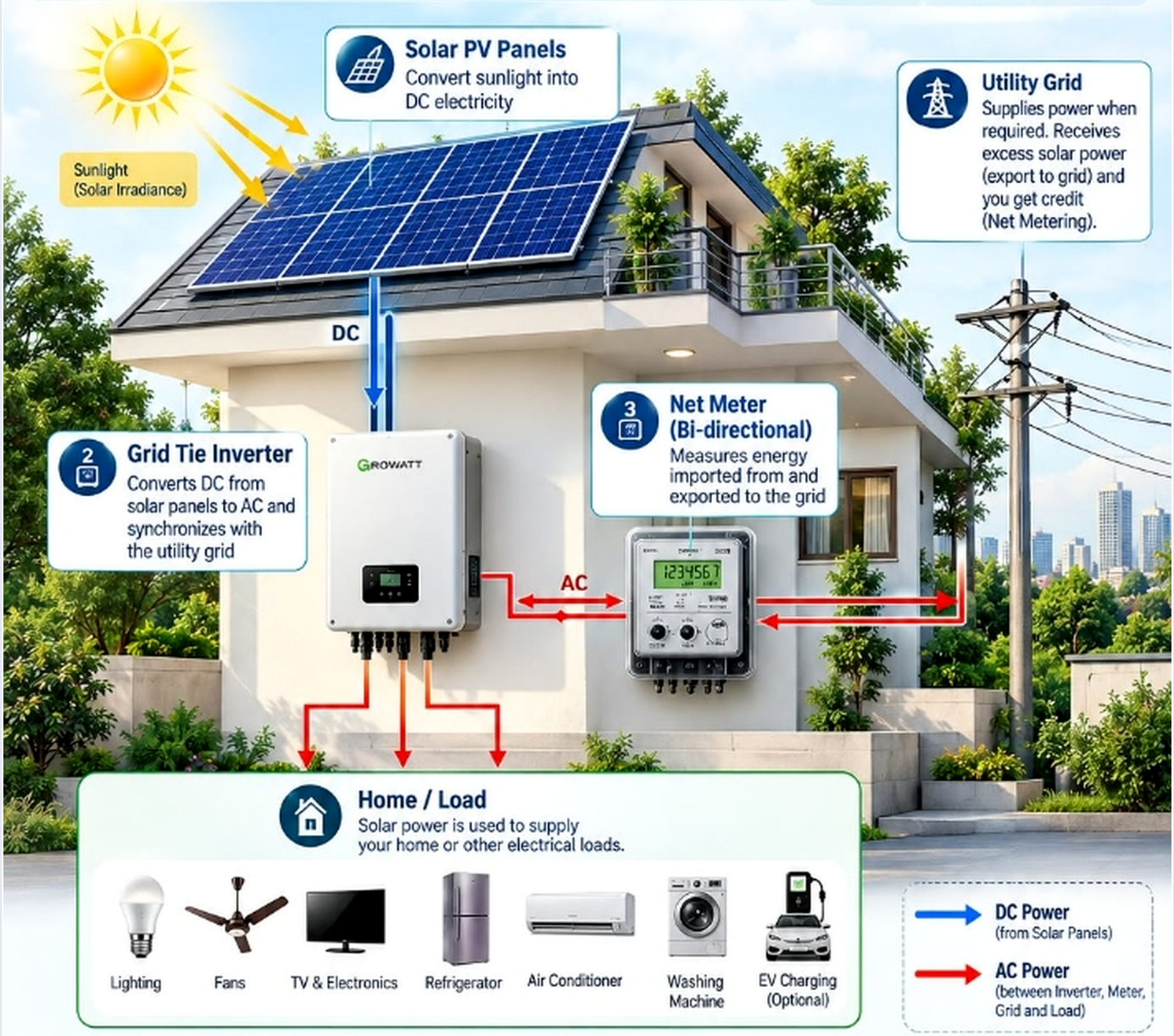
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

*Harness The Sun For A Better Tomorrow*

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future




## KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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*Clean Energy Better Tomorrow*